SNEHA GHOSH

Kolkata, West Bengal, India snehanew2003@gmail.com (+91) 9832178973 Personal Portfolio
Sneha Ghosh SnehaGhoshNew2003

OVERVIEW

As a final-year B.Tech student, I am deeply committed to advancing my expertise in machine learning, deep learning, and artificial intelligence. With a strong foundation in data science and hands-on experience in developing analytical solutions, I am eager to apply my knowledge to real-world challenges. My passion for innovation drives me to explore emerging technologies, optimize complex systems, and contribute to transformative projects in the tech industry. Actively seeking AI/ML internships or entry-level roles to contribute to impactful product development.

EDUCATION

College of Engineering and Management, Kolaghat *2021 – 2025*
Bachelor of Technology in Computer Science and Engineering
CGPA: 8.21

Kendriya Vidyalaya Ranaghat *2020 – 2021*
Class XII, Science
Percentage: 87.4

Kendriya Vidyalaya Ranaghat *2018 – 2019*
Class X, Science
Percentage: 85.8

SKILLS

Technical Skills : Python, SQL, Statistics, Probability, Linear Algebra, Calculus

Technical Tools : Power BI, Excel, Numpy, Pandas, Matplotlib, Scipy(Beginner), statsmodels(Beginner), Seaborn, Tensorflow, Keras, Langchain

Soft Skills : Adaptability, Growth Mindset, Open to feedbacks, Communication Skills

CERTIFICATIONS

Udemy: The Data Science Course – Complete Data Science Bootcamp (2025)
Key learnings:

- Data preprocessing and feature engineering.
- Mathematical foundations of machine learning.
- Implemented linear and logistic regression models using Python.
- Built ML algorithms with `NumPy`, `statsmodels`, and `scikit-learn`.
- Applied deep learning using `TensorFlow`.
- Techniques to handle underfitting, overfitting, cross-validation, and hyperparameter tuning.

TCS iON: IIT Kharagpur AI4ICPS Certificate Programme
Highlights:

- AI applications in real-world industry scenarios.
- Concepts in regression, classification, and ML mathematics.
- Practical use of `NumPy` and `pandas`.
- Studied ML algorithms: SVM, KNN, Naïve Bayes, clustering.
- Explored neural networks, computer vision, transformers, and transfer learning.
- Fine-tuning and hyperparameter optimization for model improvement.

PROJECTS

ATS Tracking System *Click here*

- Developed an **AI-powered Applicant Tracking System** to evaluate resumes against job descriptions.
- Integrated **Google Gemini API** to analyze resumes and assess job role compatibility.
- Implemented **PDF processing** using `pdf2image` and `PIL` to extract resume content.
- Provided **career development insights** with **personalized skill improvement** suggestions.
- Calculated **resume-job match percentage** and **highlighted missing keywords** for ATS optimization.
- Designed an **interactive** and **user-friendly** interface using `Streamlit` for real-time feedback.

Sentiment Analysis *Click here*

- Developed a deep learning model to analyze and classify sentiment from text reviews.
- Preprocessed data using `NLTK`, `Tokenizer`, and `pad_sequences` for **efficient** text handling.
- Implemented `LSTM` with word embeddings for improved sentiment prediction accuracy.
- Optimized model performance through **hyperparameter tuning** and **dropout** regularization.

Youtube Transcriber *Click here*

- Developed a YouTube Transcriber that **extracts** and **converts** video speech into text.
- Implemented **YouTube API** to fetch video content and process audio streams.
- Designed an **interactive** user interface using Streamlit for seamless transcription requests.
- Enabled multi-language support for transcribing videos in different languages.
- Integrated **text summarization** to generate **concise summaries** of video content.

Image Captioning *Click here*

- Developed a deep learning model to generate descriptive captions for images.
- Implemented a **CNN-RNN** `Encode-Decoder` architecture using `Xception` for **feature extraction** and `LSTM` for **caption generation**.
- Utilized the `Flickr8k` dataset for **training** and **validation**.
- Integrated `TensorFlow` and `Keras` to **build** and **optimize** the model.

Yoga Pose Detection and Prediction *Click here*

- Developed a deep learning-based system to detect and classify yoga poses from images and video frames.
- Implemented `YOLO` for **real-time** body part detection and pose localization.
- Utilized `VGG16` with transfer learning to accurately classify various yoga poses.
- Applied **pose estimation** techniques to analyze and predict body posture.
- Designed an interactive visualization for **pose correction** and **feedback**.
- Optimized model performance to ensure real-time processing efficiency.

DASHBOARDS

Hospitality Domain: Created a Hospitality Domain Analysis using Power BI [**See it here**]

Sales Analysis: Created a Sales Insight using Power BI. [**See it here**]